EE 533 Information Theory

Week 10

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A Review of CDF and continuous random variables

Let $\mathsf{X} : \Omega \to \mathbb{R}$ be a real valued random variable.

- The *cumulative distribution function* $F_{\mathsf{X}} : \mathbb{R} \to [0, 1]$

$$F_{\mathsf{X}}(x) := \mathbf{P}(\mathsf{X} \leq x) \qquad \forall x \in \mathbb{R}.$$

- CDF F_{X} is always defined.
  - If $F_{\mathsf{X}} : \mathbb{R} \to [0, 1]$ is piecewise constant, then X is a *discrete random variable*. (Converse statement is not true.)
  - X is a *continuous random variable* iff $F_{\mathsf{X}} : \mathbb{R} \to [0, 1]$ is continuous.
  - X is an *absolutely continuous random variable* iff there exists a *probability density function* $f_{\mathsf{X}} : \mathbb{R} \to \mathbb{R}_{\geq 0}$ satisfying

$$F_{\mathsf{X}}(x) := \int_{-\infty}^{x} f_{\mathsf{X}}(\tau)\, \mathrm{d}\tau \qquad \forall x \in \mathbb{R}.$$

  - X is a singular continuous random variable if $F_{\mathsf{X}} : \mathbb{R} \to [0, 1]$ is continuous and there exists Lebesgue measure, i.e. length, zero set $\mathcal{A}$ such that $\mathbf{P}(\mathsf{X} \in \mathcal{A}) = 1$, e.g. Cantor distribution.
- It is possible to extend the notion of PDF via Dirac delta function $\delta(\cdot)$.
  (i) $\delta(x) = 0$ for all $x \neq 0$.
  (ii) $\int_{-\epsilon}^{\epsilon} g(x)\delta(x)\mathrm{d}x = g(0)$ for all $g : \mathbb{R} \to \mathbb{R}$ continuous at $x = 0$.

  These extensions are beyond the scope of our discussion.

Differential entropy

The *differential entropy* $h(\mathsf{X})$ of an absolutely continuous random variable X is

$$h(\mathsf{X}) := \int f_{\mathsf{X}}(x) \log \frac{1}{f_{\mathsf{X}}(x)} \mathrm{d}x,$$
$$= \mathbf{E}\left[\log \frac{1}{f_{\mathsf{X}}(\mathsf{X})}\right]$$

- As was the case in the definition of entropy, $0 \log 0 = 0$ by convention.
- $\int g(x)\mathrm{d}x$ is defined iff either $\int \max\{g(x), 0\}\mathrm{d}x$ or $\int \max\{-g(x), 0\}\mathrm{d}x$ is finite.
- $\mathbf{E}[\mathsf{Y}]$ is defined iff either $\mathbf{E}\left[|\mathsf{Y}|^+\right]$ or $\mathbf{E}\left[|-\mathsf{Y}|^+\right]$ is finite, where $|z|^+ = \max\{z, 0\}$.
- $h(\mathsf{X})$ can be interpreted as a function of f_{X}. Hence $h(\mathsf{X})$ is sometimes denoted as $h(f_{\mathsf{X}})$.
- Recall the definition of entropy and the analogous expression in term of PMF

$$H(\mathsf{X}) := \sum_x p_{\mathsf{X}}(x) \log \frac{1}{p_{\mathsf{X}}(x)}$$
$$= \mathbf{E}\left[\log \frac{1}{p_{\mathsf{X}}(\mathsf{X})}\right]$$

- Unlike the entropy the differential entropy can be negative!

Differential entropy can take any real value

- **Uniform Distribution:** $f_X(x) = \frac{1}{\ell}\mathbb{1}_{\{x\in[0,\ell]\}}$ for $\ell > 0$.

$$h(X) = \log \ell$$

- **Exponential Distribution:** $f_Y(y) = \lambda e^{-\lambda y}\mathbb{1}_{\{y\geq 0\}}$ for $\lambda > 0$.

$$\begin{aligned} h(Y) &= \mathbf{E}\left[\log \frac{1}{\lambda e^{-\lambda Y}}\right] \\ &= -\log\lambda + \lambda \mathbf{E}[Y]\log e \\ &= \log {}^{e}/_{\lambda} \end{aligned}$$

- **Gaussian Distribution:** $f_Z(z) = \frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{(z-\mu)^2}{2\sigma^2}}$ for $\sigma > 0$ and $\mu \in \mathbb{R}$.

$$\begin{aligned} h(Z) &= \mathbf{E}\left[\log\sqrt{2\pi\sigma^2}e^{\frac{(Z-\mu)^2}{2\sigma^2}}\right] \\ &= \log\sqrt{2\pi\sigma^2} + \frac{\log e}{2\sigma^2}\mathbf{E}\left[(Z-\mu)^2\right] \\ &= \frac{1}{2}\log 2\pi\sigma^2 e \end{aligned}$$

- $h(X)$, $h(Y)$, $h(Z)$ can take any real value depending on ℓ, λ, σ.

Differential entropy can be ∞

$$F_X(x) = 1 - \frac{1}{\ln(x \vee e)}$$

$$f_X(x) = \begin{cases} \frac{1}{x \ln^2 x} & x > e \\ 0 & x \leq e \end{cases}$$

$$h(X) = \int_e^\infty \left[\frac{1}{x \ln^2 x} \log(x \ln^2 x)\right] \mathrm{d}x$$

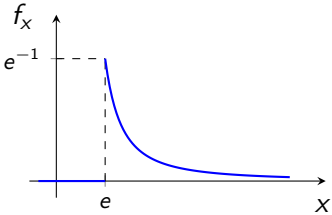
$$\geq \int_e^\infty \left[\frac{1}{x \ln^2 x} \log x\right] \mathrm{d}x \qquad \text{because } \ln^2 x \geq 1 \quad \forall x > e$$

$$= \int_e^\infty \frac{\log e}{x \ln x} \mathrm{d}x \qquad \text{because } \ln x = \frac{\log x}{\log e}$$

$$= \int_1^\infty \frac{\log e}{z} \mathrm{d}z \qquad \text{for } z = \ln x$$

$$= \log(z)|_1^\infty$$

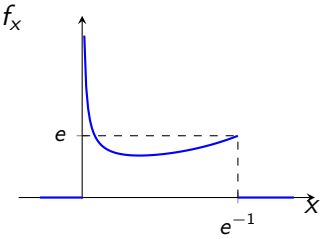
$$= \infty$$



Differential entropy can be $-\infty$

$$F_X(x) = -\frac{1}{\ln((0\vee x)\wedge e^{-1})}$$

$$f_X(x) = \begin{cases} \frac{1}{x\ln^2 x} & x \in (0, e^{-1}) \\ 0 & x \notin (0, e^{-1}) \end{cases}$$



$$\begin{aligned}
h(X) &= \int_0^{e^{-1}} \left[\frac{1}{x\ln^2 x}\log(x\ln^2 x)\right] \mathrm{d}x \\
&= \int_0^{e^{-1}} \frac{1}{x\ln^2 x}\left[\tfrac{1}{2}\log x + \tfrac{1}{2}\log(x\ln^4 x)\right] \mathrm{d}x \\
&\le \tfrac{1}{2}\int_0^{e^{-1}} \frac{\log e}{x\ln x}\mathrm{d}x + \tfrac{1}{2}\int_0^{e^{-1}} \frac{1}{x\ln^2 x}\log\left(\tfrac{4}{e}\right)^4 \mathrm{d}x && \text{because } \sup_{x\in(0,1)} x\ln^4 x = (4/e)^4 \\
&= \tfrac{1}{2}\int_\infty^1 \frac{\log e}{z}\mathrm{d}z + 2\log\tfrac{4}{e} && \text{for } z = -\ln x \\
&= \tfrac{1}{2}\log(z)\Big|_\infty^1 + 2\log\tfrac{4}{e} \\
&= -\infty
\end{aligned}$$

Differential entropy is not defined for some absolutely continuous r.v.

- If $f_X(x) = \begin{cases} \frac{1}{x \ln^2 x} & x \in (e, \infty) \\ 0 & x \notin (e, \infty) \end{cases}$, then $h(X) = \infty$
- If $f_Y(y) = \begin{cases} \frac{1}{y \ln^2 y} & y \in (0, e^{-1}) \\ 0 & y \notin (0, e^{-1}) \end{cases}$, then $h(Y) = -\infty$
- If $f_Z(z) = \begin{cases} \frac{1}{2z \ln^2 z} & z \in (0, e^{-1}) \cup (e, \infty) \\ 0 & z \in (-\infty, 0] \cup [e^{-1}, e] \end{cases}$, then $h(Z) = ?$
- $h(Z)$ is undefined because $\mathbf{E}\left[\log \frac{1}{f_Z(Z)}\right]$ is undefined. That is, neither $\mathbf{E}\left[\left|\log \frac{1}{f_Z(Z)}\right|^+\right]$, nor $\mathbf{E}\left[\left|-\log \frac{1}{f_Z(Z)}\right|^+\right]$ is finite.

Remark

The entropy is defined for all discrete random variables, i.e. $H(X)$ is defined whenever p_X is.

Differential entropy and one-to-one differentiable transforms

Lemma

Let $g : \mathcal{A} \to \mathbb{R}$ be invertible and differentiable, X be an absolutely continuous r.v. such that $\mathbf{P}(\mathsf{X} \in \mathcal{A}) = 1$, and $\mathsf{Y} = g(\mathsf{X})$, then $h(\mathsf{Y}) = h(\mathsf{X}) + \mathbf{E}\left[\log |g'(\mathsf{X})|\right]$.

Proof.

$$
\begin{aligned}
h(\mathsf{Y}) &= \mathbf{E}\left[\log \tfrac{1}{f_{\mathsf{Y}}(\mathsf{Y})}\right] \\
&= \mathbf{E}\left[\log \left|g'(g^{-1}(\mathsf{Y}))\right| + \log \tfrac{1}{f_{\mathsf{X}}(g^{-1}(\mathsf{Y}))}\right] && \text{because } f_{\mathsf{Y}}(\tau) = \tfrac{1}{|g'(g^{-1}(\tau))|} f_{\mathsf{X}}\left(g^{-1}(\tau)\right) \\
&= \mathbf{E}\left[\log \left|g'(\mathsf{X})\right| + \log \tfrac{1}{f_{\mathsf{X}}(\mathsf{X})}\right] && \text{because } \mathsf{X} = g^{-1}(\mathsf{Y}) \\
&= \mathbf{E}\left[\log \left|g'(\mathsf{X})\right|\right] + h(\mathsf{X})
\end{aligned}
$$

□

- If $\mathsf{Y} = a\mathsf{X} + b$ for an $|a| > 0$ and $b \in \mathbb{R}$, then $h(\mathsf{Y}) = h(\mathsf{X}) + \log |a|$.
- If $\mathsf{Y} = \mathsf{X} + b$ for a $b \in \mathbb{R}$, then $h(\mathsf{Y}) = h(\mathsf{X})$.
- If $\mathsf{Y} = g(\mathsf{X})$ for an invertible function g then $p_{\mathsf{Y}}(\tau) = p_{\mathsf{X}}\left(g^{-1}(\tau)\right)$, and thus $H(\mathsf{Y}) = H(\mathsf{X})$.

Asymptotic equipartition property for absolutely continuous r.v.'s

Theorem

Let $X_1, X_2, \ldots$ be i.i.d. r.v.'s with the PDF f_X such that $|h(X)| < \infty$, then

$$\frac{1}{n} \log \frac{1}{f_{X_1^n}(X_1^n)} \xrightarrow{\mathbf{P}} h(X)$$

Proof.

If $Y_k := \log \frac{1}{f_X(X_k)}$ for all $k \in \mathbb{Z}_+$ then

- $Y_1, Y_2, \ldots$ are i.i.d. random variables.
- $\mathbf{E}[Y_k] = h(X)$ for all k by definition.
- $\mathbf{E}[|Y_k|] = \mathbf{E}[|\log f_X(X)|] < \infty$ for all k because $|h(X)| < \infty$.
- $\frac{1}{n} \sum_{k=1}^n Y_k \xrightarrow{\mathbf{P}} h(X)$ by the weak law of large numbers.
- $\sum_{k=1}^n Y_k = \log \frac{1}{f_{X_1^n}(X_1^n)}$ because $f_{X_1^n}(X_1^n) = \prod_{k=1}^n f_X(X_k)$.

□

$\mathbf{P}\left(h(X) - \epsilon \leq -\frac{1}{n} \log f_{X_1^n}(X_1^n) \leq h(X) + \epsilon\right) \to 1$ as $n \to \infty$ for all $\epsilon > 0$.

$\mathbf{P}\left(2^{-n(h(X)+\epsilon)} \leq f_{X_1^n}(X_1^n) \leq 2^{-n(h(X)-\epsilon)}\right) \to 1$ as $n \to \infty$ for all $\epsilon > 0$.

$$\mathbf{P}\left(2^{-n(h(\mathsf{X})+\epsilon)} \leq f_{\mathsf{X}_1^n}(\mathsf{X}_1^n) \leq 2^{-n(h(\mathsf{X})-\epsilon)}\right) \to 1 \text{ as } n \to \infty \text{ for all } \epsilon > 0.$$

Definition (Typical Set $\mathcal{A}_\epsilon^{(n)}$ and Volume)

$$\mathcal{A}_\epsilon^{(n)} := \left\{ x_1^n \in \mathbb{R}^n : 2^{-n(h(\mathsf{X})+\epsilon)} \leq f_{\mathsf{X}_1^n}(x_1^n) \leq 2^{-n(h(\mathsf{X})-\epsilon)} \right\} \qquad \forall n \in \mathbb{Z}_+, \epsilon > 0 \qquad (1)$$

$$vol\,(\mathcal{A}) := \int_{\mathcal{A}} \mathrm{d}x_1^n \qquad \mathcal{A} \subset \mathbb{R}^n \qquad (2)$$

$$\mathbf{P}(\mathsf{X}_1^n \in \mathcal{A}) = \int_{\mathcal{A}} f_{\mathsf{X}_1^n}(x_1^n)\,\mathrm{d}x_1^n \qquad \mathcal{A} \subset \mathbb{R}^n \qquad (3)$$

Theorem ([CT06, Thm 8.2.2])

(a) $\forall \delta > 0$, $\mathbf{P}\left(\mathsf{X}_1^n \in \mathcal{A}_\epsilon^{(n)}\right) \geq 1 - \delta$ *for sufficiently large n* — *AEP*

(b) $vol\left(\mathcal{A}_\epsilon^{(n)}\right) \leq 2^{n(h(\mathsf{X})+\epsilon)}$ — (1), (2), (3), & $\mathbf{P}\left(\mathsf{X}_1^n \in \mathcal{A}_\epsilon^{(n)}\right) \leq 1$

(c) $\forall \delta > 0$, $vol\left(\mathcal{A}_\epsilon^{(n)}\right) \geq (1-\delta)2^{n(h(\mathsf{X})-\epsilon)}$ *for sufficiently large n* — (1), (2), (3), & *part* (a)

(d) $\left|\frac{1}{n}\log vol\left(\mathcal{A}_\epsilon^{(n)}\right) - h(\mathsf{X})\right| \leq 2\epsilon$ *for all n large enough.* — *part (b), (c)*

Is $\mathcal{A}_\epsilon^{(n)}$ the smallest volume set that has probability at least $1-\delta$? No. But the volume of the smallest set increases with the exponent $h(\mathsf{X})$, as well.

Definition

$\mathcal{B}_\delta^{(n)}$ is the smallest volume set that has probability at least $1-\delta$, i.e. $\mathbf{P}\left(\mathsf{X}_1^n \in \mathcal{B}_\delta^{(n)}\right) \geq 1-\delta$.

- $vol\left(\mathcal{B}_\delta^{(n)}\right) \leq vol\left(\mathcal{A}_\epsilon^{(n)}\right)$ for large enough n because $\mathbf{P}\left(\mathsf{X}_1^n \in \mathcal{A}_\epsilon^{(n)}\right) \geq 1-\delta$ for large enough n.
- $\mathbf{P}\left(\mathsf{X}_1^n \in \mathcal{A}_\epsilon^{(n)}\right) \geq 1-\frac{1-\delta}{2}$ for large enough n, hence

$$\begin{aligned}
\mathbf{P}\left(\mathsf{X}_1^n \notin \mathcal{A}_\epsilon^{(n)} \cap \mathcal{B}_\delta^{(n)}\right) &\leq \mathbf{P}\left(\mathsf{X}_1^n \notin \mathcal{A}_\epsilon^{(n)}\right) + \mathbf{P}\left(\mathsf{X}_1^n \notin \mathcal{B}_\delta^{(n)}\right) && \text{by the union bound} \\
&\leq \tfrac{1-\delta}{2} + \delta \\
&= \tfrac{1+\delta}{2}.
\end{aligned}$$

Thus $\mathbf{P}\left(\mathsf{X}_1^n \in \mathcal{A}_\epsilon^{(n)} \cap \mathcal{B}_\delta^{(n)}\right) \geq \frac{1-\delta}{2}$. Hence

$$vol\left(\mathcal{A}_\epsilon^{(n)} \cap \mathcal{B}_\delta^{(n)}\right) \geq \tfrac{1-\delta}{2} e^{n(h(\mathsf{X})-\epsilon)} \qquad \text{for large enough } n.$$

- $\lim_{n\to\infty} \frac{\log vol\left(\mathcal{B}_\delta^{(n)}\right)}{n} = h(\mathsf{X})$ because $\forall \epsilon > 0$, $\left|\frac{\log vol\left(\mathcal{B}_\delta^{(n)}\right)}{n} - h(\mathsf{X})\right| \leq 2\epsilon$ for large enough n.

Differential Entropy vs Entropy

Let Y be $\mathrm{Y} := \lfloor \mathrm{X}/\Delta \rfloor$ for a continuous random variable X and $\Delta > 0$. Then

$$p_{\mathrm{Y}}(n) = \int_{n\Delta}^{(n+1)\Delta} f_{\mathrm{X}}(x)\,\mathrm{d}x. \qquad (*)$$

$$\begin{aligned}
H(\mathrm{Y}) &= \mathbf{E}\left[\log \frac{1}{p_{\mathrm{Y}}(\mathrm{Y})}\right] && \text{by definition of } H(\cdot). \\
&= \mathbf{E}\left[\log \frac{1}{p_{\mathrm{Y}}(\lfloor \mathrm{X}/\Delta \rfloor)}\right] && \text{by } (*) \\
&= h(\mathrm{X}) + \mathbf{E}\left[\log \frac{f_{\mathrm{X}}(\mathrm{X})}{p_{\mathrm{Y}}(\lfloor \mathrm{X}/\Delta \rfloor)}\right] && \text{by definition of } h(\cdot). \\
&= h(\mathrm{X}) - \log \Delta + \mathbf{E}\left[\log \frac{f_{\mathrm{X}}(\mathrm{X})\Delta}{p_{\mathrm{Y}}(\lfloor \mathrm{X}/\Delta \rfloor)}\right] && (**) \\
&\geq h(\mathrm{X}) - \log \Delta + \mathbf{E}\left[1 - \frac{p_{\mathrm{Y}}(\lfloor \mathrm{X}/\Delta \rfloor)}{f_{\mathrm{X}}(\mathrm{X})\Delta}\right] \log e && \text{by } \log \tfrac{1}{\tau} \geq [1-\tau]\log e \\
&\geq h(\mathrm{X}) - \log \Delta && (***)
\end{aligned}$$

Differential Entropy vs Entropy cont'd

$$h(X) = H(Y) + \log \Delta - \mathbf{E}\left[\log \frac{f_X(X)\Delta}{p_Y(\lfloor X/\Delta \rfloor)}\right] \qquad \text{by } (**)$$

$$\leq H(Y) + \log \Delta \qquad \text{by } (***)$$

If $f_X(x)$ is uniformly continuous then

$$\lim_{\Delta \to 0} \mathbf{E}\left[\log \frac{f_X(X)\Delta}{p_Y(\lfloor X/\Delta \rfloor)}\right] = 0 \qquad (****)$$

(****) holds under much milder conditions.

Theorem ([CT06, 8.3.1])

If f_X is a Rieman integrable function then

$$\lim_{\Delta \to 0} H\left(\lfloor \frac{X}{\Delta} \rfloor\right) + \log \Delta = h(X).$$

Thus, the entropy of an n-bit quantization of a continuous random variable X is approximately $h(X) + n$

Joint and Conditional Differential Entropy

The *joint differential entropy* $h(\mathrm{X},\mathrm{Y})$ of random variables X,Y is

$$h(\mathrm{X},\mathrm{Y}) := \iint f_{\mathrm{X},\mathrm{Y}}(x,y)\log\frac{1}{f_{\mathrm{X},\mathrm{Y}}(x,y)}\mathrm{d}x\mathrm{d}y, \qquad = \mathbf{E}\left[\log\frac{1}{f_{\mathrm{X},\mathrm{Y}}(\mathrm{X},\mathrm{Y})}\right].$$

The *condition differential entropy* $h(\mathrm{X}|\mathrm{Y})$ of X given Y is

$$h(\mathrm{X}|\mathrm{Y}) := \iint f_{\mathrm{X},\mathrm{Y}}(x,y)\log\frac{1}{f_{\mathrm{X}|\mathrm{Y}}(x|y)}\mathrm{d}x\mathrm{d}y, \qquad = \mathbf{E}\left[\log\frac{1}{f_{\mathrm{X}|\mathrm{Y}}(\mathrm{X}|\mathrm{Y})}\right]$$

- The existence of f_{X} and f_{Y} does not ensure the existence of either $f_{\mathrm{X},\mathrm{Y}}$ or $f_{\mathrm{X}|\mathrm{Y}}$, e.g. $\mathrm{Y}=1+\mathrm{X}$.
- $h(\mathrm{X}|\mathrm{Y})$ is defined for the case when Y is a discrete random variable through the expectation.
- $\mathbf{E}[\mathrm{X}|\mathrm{Y}]$ is a random variable $h(\mathrm{X}|\mathrm{Y})$ is not!

$$h(\mathrm{X}|\mathrm{Y}=y) := \int f_{\mathrm{X}|\mathrm{Y}}(x|y)\log\frac{1}{f_{\mathrm{X}|\mathrm{Y}}(x|y)}\mathrm{d}x, \qquad = \mathbf{E}\left[\log\frac{1}{f_{\mathrm{X}|\mathrm{Y}}(\mathrm{X}|\mathrm{Y})}\middle|\mathrm{Y}=y\right].$$

- The chain rule for the differential entropy follows from $f_{\mathrm{X},\mathrm{Y}} = f_{\mathrm{X}|\mathrm{Y}}f_{\mathrm{Y}} = f_{\mathrm{X}}f_{\mathrm{Y}|\mathrm{X}}$,

$$h(\mathrm{X},\mathrm{Y}) = h(\mathrm{X}|\mathrm{Y}) + h(\mathrm{Y}) \qquad = h(\mathrm{X}) + h(\mathrm{Y}|\mathrm{X})$$

- If $\mathrm{Z} = A\mathrm{X} + b$ for an invertible matrix A then $h(\mathrm{Z}) = h(\mathrm{X}) + \log|det(A)|$ because $f_{\mathrm{Z}}(z) = A^{-1}f_{\mathrm{X}}\left(A^{-1}(x-b)\right)$.

Multivariate Normal Distribution

For random variables $X_1, X_2, \ldots, X_n$ mean μ and covariance matrix K are

$$\mu := \mathbf{E}\left[\overrightarrow{X}\right] \qquad \Rightarrow \qquad \mu_i = \mathbf{E}\left[X_i\right]$$

$$K := \mathbf{E}\left[(\overrightarrow{X} - \mu)(\overrightarrow{X} - \mu)^T\right] \qquad \Rightarrow \qquad K_{i,j} = \mathbf{E}\left[(X_i - \mu_i)(X_j - \mu_j)\right]$$

where

$\overrightarrow{X}$ short hand for the column matrix $\begin{bmatrix} X_1 \\ \vdots \\ X_n \end{bmatrix}$

A^T transpose of A, e.g. $(\overrightarrow{X})^T = \begin{bmatrix} X_1 & \cdots & X_n \end{bmatrix}$

$$\overrightarrow{X} \sim \mathcal{N}(\mu, K) \text{ iff } f_{\overrightarrow{X}}(x) = \frac{1}{(2\pi)^{n/2} det(K)^{1/2}} e^{-\frac{1}{2}(x-\mu)^T K^{-1}(x-\mu)}$$

where

$det\,(K)$ determinant of the covariance matrix K

K^{-1} is the inverse of the matrix covariance matrix K

Differential entropy of the multivariate normal distribution

Lemma

If $\overrightarrow{X} \sim \mathcal{N}(\mu, K)$, *then* $h\left(\overrightarrow{X}\right) = \log\sqrt{(2\pi e)^n det\,(K)}$.

Proof.

$$
\begin{aligned}
h\left(\overrightarrow{X}\right) &= \mathbf{E}\left[-\log\frac{1}{(2\pi)^{n/2}det(K)^{1/2}}e^{-\frac{1}{2}(\overrightarrow{X}-\mu)^T K^{-1}(\overrightarrow{X}-\mu)}\right] \\
&= \frac{\log(2\pi)^n det(K)}{2} + \frac{1}{2}\mathbf{E}\left[(\overrightarrow{X}-\mu)^T K^{-1}(\overrightarrow{X}-\mu)\right]\log e \\
&= \frac{\log(2\pi)^n det(K)}{2} + \frac{1}{2}\sum_{i,j}\mathbf{E}\left[(X_i-\mu_i)(K^{-1})_{i,j}(X_j-\mu_j)\right]\log e \\
&= \frac{\log(2\pi)^n det(K)}{2} + \frac{1}{2}\sum_{i,j}(K^{-1})_{i,j}\mathbf{E}\left[(X_j-\mu_j)(X_i-\mu_i)\right]\log e \\
&= \frac{\log(2\pi)^n det(K)}{2} + \frac{1}{2}\sum_{i,j}(K^{-1})_{i,j}K_{j,i}\log e \\
&= \frac{\log(2\pi)^n det(K)}{2} + \frac{1}{2}\sum_{i}(K^{-1}K)_{i,i}\log e \\
&= \frac{\log(2\pi)^n det(K)}{2} + \frac{1}{2}\sum_{i} I_{i,i}\log e \qquad \text{by } K^{-1}K = I \\
&= \frac{\log(2\pi)^n det(K)}{2} + \frac{n}{2}\log e
\end{aligned}
$$

Theorem

a) $h(f_X)$ *is a strictly concave function of* f_X.

b) $h(X|Y) \leq h(X)$. *The inequality is strict unless* X *and* Y *are independent.*

Proof.

a)
$$\begin{aligned}
\beta h(f_{X_1}) + (1-\beta) h(f_{X_2}) &= \int [\beta g(f_{X_1}(x)) + (1-\beta) g(f_{X_2}(x))]\,\mathrm{d}x && \text{for } g(z) := z \log \tfrac{1}{z} \\
&\leq \int g(\beta f_{X_1}(x) + (1-\beta) f_{X_2}(x))\mathrm{d}x && \text{by Jensen's ineq} \\
&= h(\beta f_{X_1} + (1-\beta) f_{X_2})
\end{aligned}$$

The inequality is strict unless $f_{X_1} = f_{X_2}$ because $g(\cdot)$ is strictly concave.

b)
$$\begin{aligned}
h(X|Y) &= \mathbf{E}\left[\mathbf{E}\left[-\log f_{X|Y}(X|Y) \middle| Y\right]\right] && \text{law of iterated expectations} \\
&= \mathbf{E}\left[h\left(f_{X|Y}(\cdot|Y)\right)\right] && \text{by the defintion of } h(\cdot) \\
&\leq h\left(\mathbf{E}\left[f_{X|Y}(\cdot|Y)\right]\right) && \text{by Jensen's ineq. and concavity of } h(\cdot). \\
&= h(X) && \text{by } \mathbf{E}\left[f_{X|Y}(x|Y)\right] = f_X(x) \text{ and } h(f_X) = h(X).
\end{aligned}$$

The inequality is strict unless $f_{X|Y}(x|y) = f_X(x)$ because $h(\cdot)$ is strictly concave.

Hadamard's inequality and Independence Bound

Lemma (Hadamard's Inequality)

$det\,(K) \leq \prod_{i=1}^{n} K_{i,i}$ for any positive definite matrix K.

Proof.

For all positive definite K, there exist X_1^n such that $X_1^n \sim \mathcal{N}(0, K)$. Thus

$$\begin{aligned}
\log \sqrt{(2\pi e)^n det\,(K)} &= h\,(X_1^n) && \text{the dif. ent. of multivariate normal r.v.'s} \\
&= h\,(X_1) + \sum_{t=2}^{n} h\,(X_t | X_1^{t-1}) && \text{by the chain rule for } h\,(\cdot) \\
&\leq \sum_{i=1}^{n} h\,(X_i) && \text{because conditioning decreases } h\,(\cdot) \\
&= \sum_{i=1}^{n} \log \sqrt{2\pi e K_{i,i}} && \text{because } X_i \sim \mathcal{N}(0, K_{i,i}) \\
&= \log \prod_{i=1}^{n} \sqrt{2\pi e K_{i,i}}
\end{aligned}$$

□

Lemma (Independence Bound)

$h\,(X_1^n) \leq \sum_{t=1}^{n} h\,(X_t)$ for jointly absolutely continuous random variable $X_1, \ldots, X_n$.

Mutual Information, Conditional Mutual Information, Chain Rule

The *mutual information* between absolutely continuous random variables X and Y is

$$\begin{aligned} I(X;Y) &:= h(X) - h(X|Y) \\ &= h(X) + h(Y) - h(X,Y) \\ &= h(Y) - h(Y|X) \end{aligned}$$

Theorem

a) $I(X;Y) = I(Y;X)$ *(definition of $I(X;Y)$)*

b) $I(X;Y) \geq 0$, *with equality iff* X *and* Y *are independent.* *($h(X|Y) \leq h(X)$)*

The *conditional mutual information* between X and Y given Z is

$$I(X;Y|Z) := h(X|Z) - h(X|Z,Y)$$

Theorem

a) $I(X;Y|Z) \geq 0$, *with equality iff* X *and* Y *are conditionally independent given* Z.

b) $I(X,Y;Z) = I(X;Z) + I(Y;Z|X)$, *i.e. the chain rule for mutual information.* *(chain rule for differential entropy)*

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